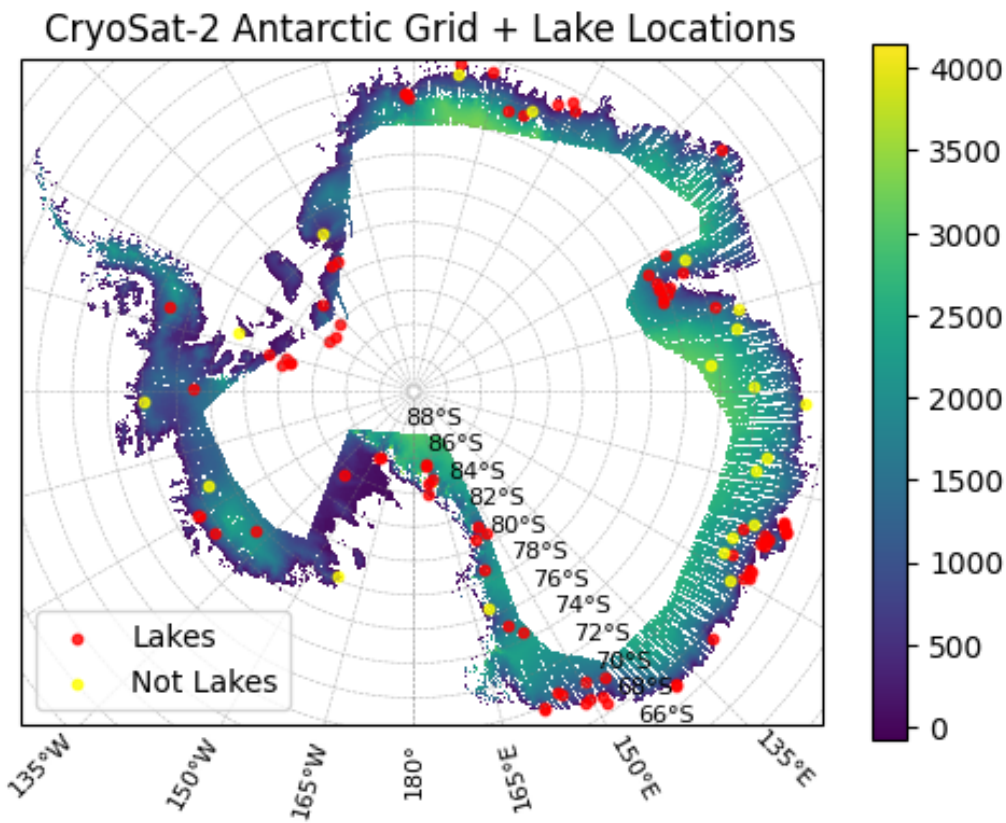
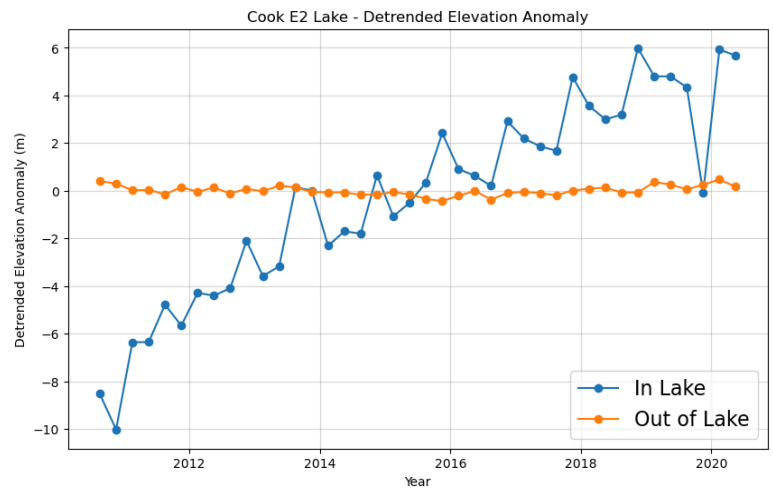
(A)


CryoSat-2 Antarctic Grid + Lake Locations
88°S
86°S
84°S
82°S
80°S
78°S
76°S
74°S
72°S
70°S
68°S
66°S
Lakes
Not Lakes
135°W
150°W
165°W
180°
165°E
150°E
135°E
4000
3500
3000
2500
2000
1500
1000
500
0

(B)


Cook E2 Lake - Detrended Elevation Anomaly
Detrended Elevation Anomaly (m)
Year
In Lake
Out of Lake
6
4
2
0
−2
−4
−6
−8
−10
2012
2014
2016
2018
2020

(C)

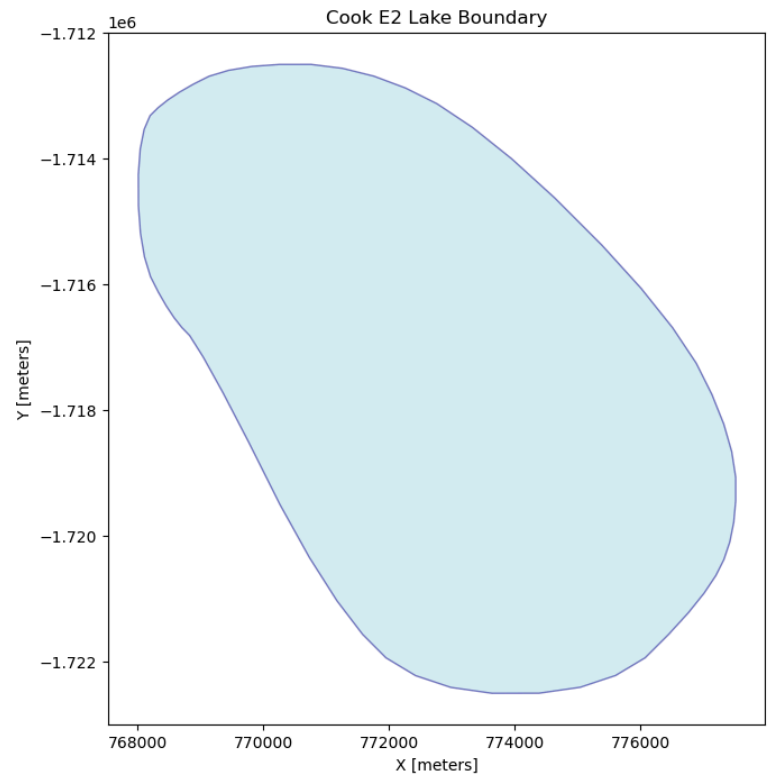

Cook E2 Lake Boundary
1e6
−1.712
−1.714
−1.716
−1.718
−1.720
−1.722
Y [meters]
768000
770000
772000
774000
776000
X [meters]

Figure 3. Panels 1 through 3 show our data and validation visualizations to determine their effectiveness for our CNN model training. The map of Antarctica visualization contains spatial distribution of subglacial lake locations in red and non lake (null) locations in yellow (Panel A). These data points could be used as training inputs for the CNN model. Time series elevation measurements at a coordinate inside of Cook lake are in blue and time series elevation measurements using a coordinate outside of the lake in orange, this was taken from the Wilson et al. study (Panel B). The data taken from within the boundary of the lake shows a clear change in elevation anomalies ranging from -10m to 6m which indicates a hydrologic filling event occurred from 2010 to 2020, while the coordinate taken from outside of the lake shows no clear elevation anomalies staying approximately at 0m. This data confirmed our training could use these lake locations and null locations to have clear parameters for validation. The shape and coordinate visualization of Cook shows the valid coordinates for pulling elevation data in the blue, while the white shows null data coordinates we could use to train our data on non lakes (Panel C). This allowed us to determine the spacing and location of our training set.